

Ayesh Sharuka

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PROFILE

AI Engineer with hands-on experience designing, building, and shipping production-grade AI systems end-to-end. Proven expertise in Python, FastAPI, and AWS (Bedrock, EC2, S3, Lambda, SageMaker), with delivered agentic systems, LLM-powered applications, RAG pipelines, cloud-native microservices, and containerised deployments. Experienced with LLM APIs (Claude, LLaMA, Gemini), agent orchestration frameworks (LangChain, MCP), and enterprise security patterns (JWT/OAuth, RBAC, CORS) as production defaults. Skilled in model evaluation, observability instrumentation, and data pipeline design with a focus on lineage and reproducibility. MSc in Artificial Intelligence at Dublin Business School, with a strong statistical foundation (BSc, University of Colombo) underpinning rigorous ML methodology. Adept at translating ambiguous business problems into scalable, maintainable AI products, owning architecture, evaluation, and documentation across the full system lifecycle.

WORK EXPERIENCE

EMUQ TECH Inc, AI/ML Developer 07/2024 – 09/2025 | United States

- Engineered a production-grade FastAPI service integrated with AWS EC2 and S3 to automate the end-to-end audio mastering pipeline, reducing manual processing time significantly.
- Built a dynamic SEO-optimised content generation platform using FastAPI and AWS Bedrock (Claude & LLAMA LLMs), generating keywords, meta descriptions, titles, and CTAs from structured business inputs.
- Designed Pydantic-validated APIs with JWT authentication and CORS controls, ensuring secure, extensible, and well-documented endpoints.

Dialog Axiata PLC, 02/2024 – 10/2024 | Sri Lanka
Industrial Training in Group Analytics & AI Platform

- Developed and deployed a Natural Language to SQL system using Google Gemini LLM, enabling non-technical users to query multi-table relational databases via natural language with measurable accuracy gains.
- Built an AWS CloudWatch resource-utilisation dashboard monitoring SageMaker pipelines; automated data ingestion via Lambda functions, saving pipeline data to S3 as CSV for downstream analytics.
- Created Python scripts to extract pipeline metrics (CPU, memory, disk, instance types) and generate custom CloudWatch widgets, contributing to cost reduction of automated ML pipelines.

Nildiya Hotel Restaurant & Bar, Data Analyst Internship 05/2023 – 02/2024 | Sri Lanka

- Collected, cleaned, and analysed operational and customer data using SQL and Python; findings directly informed service quality improvements and management reporting.
- Performed exploratory data analysis (EDA) on customer feedback datasets, identifying actionable trends that influenced strategic planning decisions.
- Produced visualisations and written reports presented to management to support data-driven decision-making.

EDUCATION

Dublin Business School, MSc, Artificial Intelligence 09/2025 – Present | Ireland

University of Colombo, 01/2020 – 05/2023 | Sri Lanka
B.Sc., Industrial Statistics & Mathematical Finance

SKILLS

Languages (Python, SQL, HTML, TypeScript) | **AI / ML** (Large Language Models (LLMs), Agentic Systems, RAG Pipelines, NLP, Deep Learning, Generative AI, Prompt Engineering, Machine Learning, Model Evaluation, Computer Vision) | **Agent & Orchestration** (LangChain, LangGraph, Tool Calling, Sub-agents, Human-in-the-Loop, MCP, Model Selection Trade-offs) | **Frameworks & Libraries** (FastAPI, Streamlit, Pydantic, Scikit-learn, TensorFlow/Keras, Pandas, NumPy, SQLite3) | **Cloud & Infrastructure** (AWS (Bedrock, EC2, S3, Lambda, SageMaker, CloudWatch, CodeCommit), Docker, Kubernetes (K8s), Serverless, CI/CD, Microservices, Event-driven Systems) | **LLM / AI APIs** (Claude API, LLaMA, Google Gemini API, AWS Bedrock, Azure OpenAI Services, OpenAI API) | **Security & Auth** (JWT Authentication, OAuth, RBAC, Secrets Management, CORS, GDPR-aware Design, SSO) | **Engineering Practices** (REST APIs, Microservices Architecture, Data Lineage & Reproducibility, Observability, Logging & Tracing, Latency & Throughput Optimisation, Error Handling, Testing, Technical Documentation) | **Data & Analytics** (EDA, Data Preprocessing, Statistical Analysis, Data Visualisation, Vector Stores, Embeddings, Feature Pipelines) | **Developer Tools** (Git, GitHub, Postman, Docker Compose) | **Soft Skills** (Cross-functional Collaboration, PoC-to-Production Delivery, Pragmatic Problem Solving, Research & Comparative Analysis)

PROJECTS

SEO-Optimised Content Generation Platform (FastAPI + AWS Bedrock)

- Built a cloud-native content generation API using FastAPI and AWS Bedrock, leveraging Claude and LLAMA LLMs to produce SEO-ready marketing copy (keywords, titles, meta descriptions, CTAs) from user-supplied business context.
- Designed structured JSON output schemas with Pydantic models, enabling reliable downstream consumption and extensibility for new content types.
- Demonstrated cloud-based LLM integration best practices: API security, error handling, response optimisation, and modular architecture.

Automated Audio Mastering Service (FastAPI + AWS)

- Implemented JWT-based authentication and CORS policies for a secure FastAPI application handling audio file upload, processing, and delivery via AWS S3.
- Integrated structured logging and error handling for S3 credential issues and file I/O failures, ensuring production reliability.

Natural Language to SQL - LLM Prototype

- Prototyped an NL-to-SQL engine using Google Gemini API and SQLite3; engineered custom prompts to handle multi-table schema awareness and edge-case query types.
- Deployed via Streamlit for rapid user testing; documented comparative analysis of AWS Bedrock and Azure OpenAI for production-grade cloud implementation.

CloudWatch Resource Utilisation Dashboard

- Automated SageMaker pipeline metric collection (CPU, memory, disk, instance type) using Python CloudWatch and SageMaker clients; persisted CSV snapshots to S3.
- Deployed custom CloudWatch dashboards via Lambda functions; project objective was cost reduction of automated ML pipeline infrastructure.

Customer Churn Prediction - ANN Model (AWS SageMaker)

- Developed an end-to-end churn prediction pipeline in SageMaker: data preprocessing, ANN model training, evaluation, and deployment, optimising for accuracy and inference performance.

References available upon request.